

Vaishnavi Hiremath

Embedded Systems Engineer: Firmware & Communication Protocols, PCB Design, Hardware Bring-Up

Dharwad / Bengaluru, India · +91 98803 31533 · vaishhiremath2004@gmail.com · [LinkedIn](#) · [GitHub](#)

Electronics & Communication engineer in defence R&D with end-to-end ownership of embedded hardware: firmware and peripheral bring-up on ESP32 and Raspberry Pi 4 over UART, SPI, I2C and I2S; custom PCB design in KiCad from schematic to fabrication-ready Gerber; board-level debugging with oscilloscope and logic analyzer. Strong analog and digital fundamentals in C, Embedded C and Python.

EXPERIENCE

DeepTech R&D Intern, SSS Defence

Sep 2025 – Present

Bengaluru, India (on-site) · Acoustic sensing & embedded hardware R&D

- Brought up a multi-channel microphone array on a Raspberry Pi 4 acoustic sensing prototype, integrating the capture chain over I2S and validating the signal path from sensor to Linux userspace.
- Wrote and debugged embedded C firmware interfacing microcontrollers with sensors and ADCs: GPIO, timers, interrupts, PWM and DMA configured for reliable real-time sampling.
- Owned custom PCB development in KiCad end to end, covering schematic capture, multi-layer layout, routing, DRC and Gerber generation, producing a fabrication-ready signal-processing board.
- Debugged prototype boards to component level with oscilloscope, logic analyzer and DMM, isolating signal-integrity, connectivity and power faults; selected the final microphone configuration after evaluating MEMS and electret options on directivity, SNR and frequency response.
- Built two complete prototype iterations end to end (soldering, assembly, firmware bring-up, signal-chain validation), working alongside hardware, PCB layout, FPGA and test teams; implemented a TDOA-based algorithm estimating direction of arrival from array inputs.
- Leading PCB R&D on a next-generation board integrating a microphone array with an FPGA on a single design (VHDL/Verilog digital front end), and contributing to low-power radar front-end R&D.

PROJECTS

Guidance Package, Guided System

In progress

RF front end · full-wave EM simulation · KiCad multi-board · digital back-end

- Radar link-budget and detection-performance analysis setting the sensing requirement for the RF front end; antenna and array design verified in full-wave electromagnetic simulation.
- Digital back-end and processing architecture for the receive chain; multi-board flight electronics (power, interconnect, layout) to fabrication-ready hardware and its measurement plan.
- Currently owning the RF front-end PCB, building up the receive chain alongside the board design.

Real-Time 3D Sound Source Localization

Final year project

Real-time 3D localization over a custom microphone array, pairing classical DOA estimation (GCC-PHAT, beamforming) with a neural network trained to separate simultaneous sources. Python.

Smart Health Monitoring System

Wearable tracking SpO₂, heart rate and temperature, publishing telemetry over MQTT; interrupt-driven sampling and deep-sleep modes cut average power draw. ESP32, MAX30102, DHT11, I2C, Wi-Fi.

Sign-Language Recognition: real-time gesture-to-text on MediaPipe hand landmarks: over 90% accuracy on a 26-sign ASL dataset, sub-200 ms inference. Python, OpenCV, TensorFlow.

TECHNICAL SKILLS

Languages	C, Embedded C, Python, MATLAB, VHDL & Verilog (basics)
Protocols	UART, SPI, I2C, I2S, GPIO, MQTT, Wi-Fi, Ethernet / TCP-IP
Peripherals	ADC, timers, interrupts, PWM, deep-sleep / low-power modes
Electronics	Analog and digital fundamentals, op-amps, signal conditioning, power rails, schematic and datasheet interpretation
Platforms	ESP32, Raspberry Pi 4 (embedded Linux), Arduino / AVR, 8051, Bharat Pi, FPGA
Hardware	KiCad (schematic, PCB layout, Gerber), PCB assembly, soldering, board bring-up, component-level debugging
Test	Oscilloscope, digital multimeter, logic analyzer, bench power supply
RF	Radar link budgets, detection performance, antenna & array design, full-wave EM simulation
Tools	Keil µVision, Lattice Diamond / Radiant, Git, Linux (CLI, SSH, networking)

EDUCATION & RECOGNITION

B.E., Electronics & Communication Engineering, KLE Institute of Technology, Hubballi

2022 – Jun 2026 · CGPA 8.96/10

Pre-University (XII), Science, Prism PU College, Dharwad · **SSLC**, KLE School, Dharwad

92% · 89%

Samsung Innovation Campus: IoT & AI/ML. RVCE: Advanced Antenna Design. Secretary, IEEE MTT-S Student Chapter; Student Volunteer Coordinator (2025–26).